25901406 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECMBG201 - FUNDAMENTALS OF MICROBIOLOGICAL ANALYSIS OF FOOD AND WATER

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Food intoxications are caused by the ingestion of _____ elaborated by microorganisms. [U] / [CO1]

   a). toxins b). pigments

   c). enzymes d). acids

2. Lactose-fermenting bacteria appear_____ on MacConkey agar. [K] / [CO2]

   a). yellow b). blue

   c). pink d). green

3. Which biosafety cabinet should be used while handling airborne pathogens? [K] / [CO2]

   a). Class I BSC b). Class II or Class III BSC

c). Laminar airflow cabinet d). Open bench

4. During staining, lipid removal occurs mainly during which step? [A] / [CO3]

a). Crystal violet b). Gram's Iodine

c). Acetone treatment d). Safranin

5. CFU stands for: [K] / [CO5]

a). Colony forming unit b). Colony function unit

c). Culture forming unit d). Cellular frequency unit

6. Which of the following colour change indicate positive result in MBRT? [K] / [CO5]

a). white to blue b). blue to white

c). white to yellow d). blue to black

7. *Vibrio cholerae* produces _____ coloured colonies on TCBS agar. [U] / [CO4]

a). pink b). green

c). yellow d). black

8. Natural habitats usually contain microorganisms in the form of: [K] / [CO4]

a). pure cultures b). mixed populations

c). single colonies d). subcultures

9. What is the typical pore size of the membrane filter used in water microbiology? [K] / [CO4]

a). 0.45 µm b). 0.85 µm

c). 1 µm d). 2 µm

10. Name the differential carbohydrate in DCA agar. [U] / [CO4]

a). Sorbitol b). Mannitol

c). Lactose d). Glucose

[1x10 = 10]

Part B

True or False questions

Answer all questions

Each question carries **1** marks

11. The pH of Nutrient Agar is adjusted to 5.5–6. [U] / [CO2]

12. The main aim of decontamination is to increase microbial growth. [U] / [CO3]

13. *Salmonella* produces H_2S on XLD agar. [K] / [CO4]

14. Safe handling of bacteria and fungi helps prevent contamination and laboratory hazards. [K] / [CO2]

15. Mixing after each transfer in serial dilution ensures uniform distribution of microorganisms. [U] / [CO5]

[1x5 = 5]

Part C

One word or very short answer questions

Answer all questions

Each question carries **1** marks

16. What kind of containers should be used for collecting samples for microbiology analysis? [U] / [CO2]

17. State why gloves must be worn while handling microbiological waste. [U] / [CO2]

18. Indicate the characteristic growth of *E. coli* strains on SMAC agar. [K] / [CO4]

19. State the purpose of sealing the room before fumigation. [U] / [CO3]

20. Name one selective broth used in confirmed coliform testing. [K] / [CO4]

21. How many medium squares are there in one large square of a haemocytometer? [K] / [CO4]

22. What are indicator organisms? [U] / [CO4]

23. State the motto of FSSAI. [K] / [CO2]

24. Name the tool used to spread inoculum on agar surface in spread plating. [K] / [CO5]

25. Name the tool used to streak bacterial cells on agar plate. [K] / [CO4]

[1x10 = 10]

Part D

Reason based short answer questions

Answer any **5** questions

Each question carries **1** marks

26. Why is it important to sterilize nutrient media before use? [A] / [CO2]

27. Staining improves visibility of bacterial cells under the microscope. Justify. [U] / [CO3]

28. Baird-Parker Agar is a selective and differential medium. Justify this statement. [U] / [CO4]

29. A sterile medium is used for setting blank before sample measurement in a spectrophotometer. Give reasons. [A] / [CO4]

30. Give reasons for rotating and rolling the swab during sample collection. [U] / [CO2]

31. What is the purpose of keeping only essential materials in the working area during experiments? [A] / [CO2]

32. Why is SS agar considered suitable for isolating enteric pathogens from mixed flora? [U] / [CO4]

33. State reasons why exposure to air accelerates fat spoilage. [U] / [CO5]

[1x5 = 5]

Part E

Short answer questions

Answer any **5** questions

Each question carries **2** marks

34. Discuss about the chemical hazards in foods. [U] / [CO1]

35. Give an account of the procedure for estimating acid value in edible oils. [U] / [CO5]

36. Explain the importance of EU food laws in international trade. [U] / [CO2]

37. Explain the terms inoculum and inoculation [U] / [CO5]

38. List the components of Plate Count Agar. Write down the preparation procedure. [K] / [CO2]

39. Describe the principle of LPCB staining. [U] / [CO3]

40. Briefly describe the principle of Presumptive Test. [U] / [CO4]

41. Write the composition of EMB agar. [K] / [CO4]

[2x5 = 10]

Part F

Long answer questions

Answer any **2** questions

Each question carries **5** marks

42. Describe the role of industrial and pharmaceutical sampling in ensuring product safety. [A] / [CO2]

43. Compare sterilization using dry heat and moist heat with suitable examples. [A] / [CO3]

44. Compare the growth characteristics of mannitol fermenters and non-fermenters on MSA medium. [An] / [CO4]

45. Differentiate between acid value and peroxide value. [An] / [CO5]

[5x2 = 10]